Auditing GeoFM Evaluation for Field-Extent Segmentation: Label Proxies, Baselines, and When Frozen Features Match Fine-tuning

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Reviewed on OpenReview: *https://openreview.net/forum?id=qRXVTe1yYp*

Abstract

Geospatial foundation-model (GeoFM) benchmarks for agricultural field-extent segmentation often rely on cropland land-cover proxies and compare against per-pixel spectral baselines. We measure how these choices affect conclusions using identical single-date Sentinel-2 pixels from six countries. Replacing polygon-derived field labels with ESA WorldCover cropland raises random-forest AUROC from 0.55–0.82 to 0.79–0.96, showing that the proxy creates a more spectrally separable task. On the actual field-extent labels, a from-scratch U-Net reaches 0.89–0.98 AUROC and has higher point AUROC than a Prithvi frozen linear probe in five of six regions. We then compare a frozen backbone and full fine-tuning using the same decoder for Prithvi and TerraMind, with ten matched seeds. At ε=0.02 AUROC, 9 of 12 region×model cells are TOST-equivalent. Kenya is the common failure case: positives cover roughly 0.1% of pixels, and full fine-tuning performs better for both models. Cross-region results are less favorable to frozen models. On AUROC and average precision (AP), the U-Net leads the full 30-transfer matrix, while an exploratory analysis excluding Kenya finds that both frozen decoders outperform their fine-tuned counterparts; no AUROC difference is detected between frozen TerraMind and the U-Net in that subset. For Prithvi, a frozen decoder trains about 110× fewer parameters than full fine-tuning. These results argue for auditing labels and baselines before interpreting GeoFM rankings, and for testing a frozen decoder before assuming that full fine-tuning will help. Our conclusions are limited to single-date, per-pixel field-extent segmentation, not parcel-boundary or instance delineation.

1 Introduction

Evaluating GeoFMs for agricultural field mapping requires two decisions that can change model rankings: what label defines a field, and what baseline represents supervised learning. Published studies sometimes use ESA WorldCover "cropland" because field polygons are scarce. Cropland and field extent are related but different targets: one is a land-cover class, while the other records whether a pixel lies inside a mapped field polygon. Comparisons also often stop at per-pixel spectral classifiers, even though a segmentation model can use spatial context.

We audit both decisions on identical Sentinel-2 pixels from six countries in Fields of The World. To isolate label choice, we score the same models and pixels against polygon-derived field extent and against WorldCover cropland. To test baseline strength, we compare per-pixel random forests and frozen GeoFM probes with a U-Net trained from scratch. We then turn to the adaptation question, using the same decoder, loss, schedule, and matched seeds for frozen and fully fine-tuned Prithvi and TerraMind backbones.

[*]OpenAI Codex and Anthropic Claude assisted with language editing, software debugging, and consistency checks; the author independently verified the work, made all scientific decisions, and takes full responsibility for it.

Label choice changes the benchmark before adaptation enters the comparison. WorldCover inflates the spectral-baseline score without changing the imagery, whereas polygon-derived field labels expose the need for a spatial baseline. On those labels, the U-Net has higher point AUROC than the Prithvi frozen probe in most regions, with India as the exception. PANGAEA (Marsocci et al., 2024) reports a similar supervised-baseline pattern on AI4SmallFarms. The contrasting comparison in the Prithvi-EO-2.0 report uses multi-temporal crop segmentation (Szwarcman et al., 2024) and therefore addresses a different task.

The adaptation comparison also changes once decoder capacity is matched. Frozen and fully fine-tuned backbones usually agree in-region; Kenya's sparse-positive setting is the shared failure case. Cross-region transfer is metric-dependent. On AUROC and AP, the U-Net leads the full 30-transfer matrix. On AUROC, frozen-over-fine-tuned gains appear only in the exploratory non-Kenya subset. The practical appeal of a frozen decoder therefore comes from low training cost and frequent in-region parity. The transfer results do not support a universal advantage.

Contributions. This paper provides a same-pixel audit of label and baseline choices in GeoFM field-extent evaluation, a controlled ten-seed comparison of frozen and fully fine-tuned backbones, and a reproducible protocol that reports both the successful regions and the sparse-positive Kenya failure case.

2 Related Work

Geospatial foundation models. Recent GeoFMs pretrain on large unlabeled satellite archives and are reused across tasks: masked-autoencoder models (SatMAE (Cong et al., 2022), Scale-MAE (Reed et al., 2023), Prithvi-EO (Szwarcman et al., 2024)), contrastive and multimodal models (CROMA (Fuller et al., 2023), Clay (Clay Foundation, 2024), TerraMind (Jakubik et al., 2025), AnySat (Astruc et al., 2024)), wavelength-conditioned models (DOFA (Xiong et al., 2024)), spectral models (SpectralGPT (Hong et al., 2024)), pixel-timeseries models (Presto (Tseng et al., 2023)), and large supervised pretraining (SatlasPretrain (Bastani et al., 2023)). Their downstream evaluations almost always fine-tune the backbone. We test that default for field-extent segmentation.

Frozen features vs. fine-tuning. Outside Earth observation, a substantial literature documents that frozen pretrained features with a light head are often competitive with full fine-tuning, and that fine-tuning can harm out-of-distribution performance. Kumar et al. (2022) show that fine-tuning distorts pretrained features and underperforms frozen linear probing under distribution shift, which is the pattern we observe across regions. Locked-feature training (Zhai et al., 2022), transfer studies (Kornblith et al., 2019; Kolesnikov et al., 2020), and the linear-probe evaluation of self-supervised models (He et al., 2022) all point in the same direction. We find evidence consistent with the same pattern for geospatial field-extent segmentation, with an explicit boundary condition and a cost analysis.

GeoFM-specific evaluation questions. Prior work establishes that frozen representations can remain competitive under distribution shift. It does not resolve how that finding translates to GeoFM benchmarks, where a frozen backbone is often given only a linear head while its fine-tuned counterpart uses a decoder. We remove this capacity difference by giving each frozen and fine-tuned pair the same decoder (2.75M parameters for Prithvi; 2.16M for TerraMind). This capacity-matched comparison tests whether a reported fine-tuning gain comes from backbone adaptation or from giving the fine-tuned model a stronger decoder. With decoder capacity matched, the difference falls within the declared equivalence margin in 9 of 12 cells (Section 5).

The available labels introduce a second geospatial question that is not captured by the in-distribution/out-of-distribution framing in Kumar et al. (2022): what task does the benchmark measure, and what happens when positives are exceptionally sparse? On the same pixels, WorldCover cropland raises the per-pixel random forest from 0.55–0.82 to 0.79–0.96 AUROC (Table 2), so a land-cover proxy changes the task being measured. Kenya marks the limit of the frozen-decoder result: at roughly 0.1% positive pixels and 155 training chips, the frozen decoder trails full fine-tuning by 0.10–0.12 AUROC for both GeoFMs and produces near-all-negative predictions at threshold 0.5 (Section 6). These conditions make the result a geospatial evaluation finding about where a frozen decoder works and fails. They do not establish a new general claim about frozen representations.

EO benchmarks and evaluation. GEO-Bench (Lacoste et al., 2023) and PANGAEA (Marsocci et al., 2024) standardize GeoFM evaluation; we use PANGAEA's results as external corroboration. Fields of The World (Kerner et al., 2024) supplies the field polygons that let us replace land-cover proxies. Spatial autocorrelation is a well-known evaluation pitfall in geospatial ML: train and test chips drawn from nearby tiles can inflate scores. We use a chip-grouped split (no chip in both partitions) and report tile-disjoint sensitivity (Appendix Table 17) to address it. Our subject is the evaluation design, specifically labels and baselines; we do not propose a new model.

3 Setup

Data. We use Sentinel-2 L2A chips paired with field polygons from Fields of The World (Kerner et al., 2024) across six countries spanning smallholder to industrial agriculture: India, Cambodia, Vietnam, Kenya, France, and the Netherlands. For each chip we form two labels on the same pixels. The TRUE label is 1 when the pixel lies inside an FTW field polygon, rasterized to the chip grid in the chip's own CRS. The PROXY label is 1 when the pixel is ESA WorldCover "cropland" (class 40, Zanaga et al., 2022). The task throughout is per-pixel binary field-extent segmentation (inside-vs-outside a field polygon); we do not evaluate boundary delineation or instance separation between adjacent parcels, and we use "field mapping" as a synonym for this per-pixel task.

Models. Non-FM baselines: a per-pixel random forest on the 12 raw S2 bands, a 5×5 windowed random forest (300-d), and a from-scratch U-Net (Ronneberger et al., 2015) (31M parameters). GeoFMs: Prithvi-EO-2.0-300M (Szwarcman et al., 2024) and TerraMind (Jakubik et al., 2025) (with Clay and AnySat for breadth), used in two modes. Frozen: backbone weights fixed, then either a linear probe or a trained decoder is fit on frozen features. Fine-tuned: backbone trained end-to-end with the same decoder.

Nomenclature. We use three terms consistently. **Frozen-probe** = backbone weights fixed, a linear classifier ($\sim 10^3$ parameters) trained on extracted per-pixel features; Table 2 reports Prithvi consistently, with the four-model breakdown in Appendix Table 7. **Frozen-decoder** = backbone weights fixed, a 4-stage convolutional decoder (2.75M parameters for Prithvi; 2.16M for TerraMind) trained on extracted feature maps; this is the apples-to-apples comparator against full fine-tuning and is the configuration used throughout Sections 5–7 unless noted. **Full-FT** = backbone and the same decoder trained jointly end-to-end. For Prithvi, this configuration has 307M trainable parameters. The three configurations can disagree: in Kenya, the best frozen probe across four FMs reports 0.853 AUROC (AnySat; separate frozen-probe sensitivity), while the ten-seed frozen-decoder on Prithvi reports 0.653; we report both with the configuration explicit and discuss the gap in Section 6.

Table 1: **Model configurations used in this paper.** Five configurations appear across the results tables; each row names the trainable component, trainable-parameter count, whether the backbone is updated, the training budget, and which tables report each configuration. Decoder counts use Prithvi/TerraMind order.

Configuration	Trainable params	Backbone updated	Optimizer / epochs	Reported in
per-pixel / windowed RF	(fitted trees)	n/a	sklearn defaults	Tables 2, 5
frozen-probe (GeoFM + linear)	$\sim 10^3$	no	logistic regression	Tables 2, 5, 7
frozen-decoder (GeoFM + 4-stage conv)	2.75M / 2.16M	no	AdamW, 80 epochs	Tables 3, 4, 8
full-FT (GeoFM + same decoder)	307M (Prithvi)	yes	AdamW, 80 epochs	Tables 3, 4, 8
U-Net (from scratch)	31M	n/a	Adam, 150 epochs	Tables 2, 3, 4

Protocol and statistics. All models are scored on the same held-out test pixels via a chip-grouped split (no chip in both train and test; overlap = 0; seed 20260514, 25% test). The main frozen-decoder, full-FT, and U-Net comparisons use a single-recipe ten-seed grid with ten matched training seeds per region and one software recipe; we report mean and a seed-paired t-interval with df = 9 for matched frozen-decoder vs. full-FT differences (the same paired analysis used in Table 3 and Appendix Table 8). Frozen-probe AUROC in Table 2 carries chip-level 95% bootstrap CIs (B=1000) recomputed over the test chips, available in five regions (Cambodia per-chip CIs not recomputed; point estimates only). Equivalence claims use two one-sided tests (TOST; Lakens, 2017) with declared margin ε=0.02 AUROC in-region; Holm correction across the 12 primary TOST p-values retains all 9 equivalent cells at α=0.05 (`ftw_inregion_equivalence.json`).

TOST margin justification. We use 0.02 AUROC as a practical tolerance, not as a universal standard for field mapping. It is less than half of the India frozen-probe-over-U-Net gap (0.042) that we interpret substantively in Section 4, and more than twice the largest per-configuration seed standard deviation outside the three failing cells (0.0085). The sensitivity analysis in Appendix Table 9 gives the result across margins from 0.005 to 0.030. The failing set is unchanged from 0.020 through 0.030. At 0.015, Prithvi–Netherlands also fails; at 0.010, TerraMind–Netherlands joins it. Both point estimates favor the frozen decoder. At 0.005, Prithvi–India and TerraMind–Cambodia also fail; their fine-tuning-favored differences are 0.004 and 0.006 AUROC.

Cross-region uncertainty and reproduction. Cross-region uncertainty is seed-paired over ten seeds, with each seed contributing its mean over the directed-transfer subset. The multiple-comparison family contains 10 directional tests: two claim families in each of the five regions with retained per-chip uncertainty. Cambodia is excluded because its per-chip CIs were not recomputed. The tests compare $\mathrm{RF}_{\mathrm{proxy}}-\mathrm{RF}_{\mathrm{true}}$ on F1 and best-FM$-\mathrm{RF}_{\mathrm{true}}$ on AUROC. All 10 survive both Benjamini–Hochberg FDR control at 0.05 and Holm correction at 0.05, with minimum z=6.54 (`ftw_multiple_comparison.json`); chip-paired Wilcoxon signed-rank confirms the India Prithvi-over-U-Net exception (p=2.4 $\times$ 10^{-4}, all 12 chunks positive). The evaluation code, metric artifacts, and reproduction commands are released. Primary aggregates recompute from the released run files, and direct raster checks verify label-to-image alignment (Appendix A).

4 Two Evaluation Choices that Change the Result

All GeoFM entries in this section are the frozen-probe configuration (Prithvi linear probe on extracted per-pixel features) unless labeled 'FT'; see Table 1. Table 2 reports, per region, the per-pixel RF and the strong models under both labels (TRUE-label AUROC unless noted).

Table 2: **Two evaluation choices (true-label AUROC; six regions).** The WorldCover proxy lifts the per-pixel RF from 0.55–0.82 to 0.79–0.96, making it foundation-model-competitive. A from-scratch U-Net on TRUE labels reaches 0.89–0.98 everywhere and is competitive with Prithvi frozen-probe in five of six regions (India excepted), so the reported "GeoFM $\gg$ baseline" gap mostly reflects the choice of baseline. The Prithvi frozen-probe column reports a single consistent model (linear probe on Prithvi-EO-2.0 features); a per-model breakdown across {Prithvi, TerraMind, Clay, AnySat} is in Appendix Table 7. Bracketed values are 95% CIs: chip-level bootstrap (B=1000) for the frozen-probe column (five regions; Cambodia point estimate only); seed-level t-interval (df = 9) for ten-seed U-Net and Prithvi full-FT. The full-FT column is end-to-end fine-tuned Prithvi. Per-model tile-disjoint sensitivity is in Appendix Table 17. The 'Prithvi frozen-probe' column is a linear probe on Prithvi-EO-2.0 features (*not* the frozen-decoder configuration used in Table 3); see Table 1 for the full configuration key.

Region	RF TRUE	RF PROXY	U-Net	Prithvi frozen-probe	Prithvi full-FT
India	0.574	0.786	0.941 [.925,.958]	0.983 [.979,.985]	0.985 [.984,.987]
Cambodia	0.550	0.900	0.948 [.947,.949]	0.924 (no CI)	0.949 [.949,.949]
Vietnam	0.643	0.911	0.964 [.961,.967]	0.929 [.925,.933]	0.958 [.958,.959]
Kenya	0.651	0.806	0.891 [.885,.897]	0.766 [.718,.813]	0.771 [.738,.804]
France	0.695	0.959	0.977 [.977,.978]	0.961 [.958,.964]	0.979 [.978,.980]
Netherlands	0.815	0.928	0.981 [.980,.983]	0.955 [.951,.958]	0.958 [.953,.963]

Choice 1: label source. The proxy raises the per-pixel RF by +0.11 to +0.35 AUROC in every region, turning a near-chance field-membership predictor into one that rivals GeoFMs. The reason is that WorldCover "cropland" is spectrally separable while true field membership is a spatial property. Across the five

regions with shipped coverage, WorldCover labels 22–70% of pixels as cropland, versus 0.1–18% true field coverage. The proxy therefore asks "is this managed land?" instead of "is this pixel in a field?"

Choice 2: baseline strength. On TRUE labels a supervised U-Net reaches 0.89–0.98 AUROC in all six regions. Compared against a single consistent FM (Prithvi frozen-probe), the U-Net has higher point AUROC in five of six regions (Vietnam, Kenya, France, Netherlands, Cambodia); India is the only GeoFM exception (Wilcoxon p=2.4 $\times$ 10^{-4} on per-chip paired chunks). For Vietnam, the U-Net's ten-seed CI $[.961, .967]$ is disjoint from frozen-Prithvi's $[.925, .933]$; for Cambodia we have only point estimates (no per-chip CI). India is the only region where frozen-Prithvi's CI $[.979, .985]$ is above the U-Net point estimate (0.941). The "GeoFM $\gg$ baseline" gap therefore depends substantially on the choice of baseline; by itself, it is not evidence of a transferable representational advantage. A per-pixel RF cannot use spatial context, and a 5×5 windowed RF closes only about 0.01 of the gap (with a small drop on Kenya; Table 15). The per-region best across {Prithvi, TerraMind, Clay, AnySat} (Appendix Table 7) raises the frozen-FM Kenya and Netherlands cells to 0.853 (AnySat) and 0.961 (AnySat), but tile-disjoint sensitivity (Appendix Table 17) shows the AnySat advantage does not survive a split-design change; on the apples-to-apples Prithvi-only frozen-probe comparison reported above, the U-Net is the stronger model in those two regions.

India exception. In-region, Prithvi frozen-probe beats the U-Net in only one of six regions (India), where Prithvi leads by 0.042 AUROC, and the direction is supported by a separate chip-paired one-sided Wilcoxon signed-rank test (n=12 paired chunks; p=2.4 $\times$ 10^{-4}; all 12 pairs positive). The chunk audit shows that India is stable across test chunks rather than a seed artifact, so we retain it as an exception. Possible contributors include Prithvi's HLS global pretraining (which may include Indian agro-climates in unknown proportion), India's distinctive smallholder spatial layout, and ordinary single-region noise at the between-region scale. The released alignment summary gives Cambodia a smaller mean polygon area than India (0.20 vs. 0.36 ha), but the U-Net leads its frozen probe (0.948 vs. 0.924), so field size alone does not explain the pattern. We do not have a single confident explanation. The India exception cautions against treating the supervised-baseline result as universal; the field-size comparisons above do not identify a mechanism.

Training-budget control. The U-Net comparison is not epoch-matched to the GeoFM full-FT runs (150 epochs from-scratch vs. 80 from a pretrained backbone). An epoch-matched U-Net control at 80 epochs is reported in Appendix Table 13. At the matched 80-epoch budget the U-Net still has higher point AUROC than the Prithvi frozen probe in five of six regions (India remains the same exception by point estimate), and the 80-epoch means are within 0.01 AUROC of the 150-epoch means in five of six regions; India shows the largest drop (0.941 $\rightarrow$ 0.911, seed std 0.023), which widens the existing India frozen-probe lead over the U-Net but does not create a new exception. Epoch count is also a poor proxy for compute here: our approximate development-run observations were $\approx$15 GPU-min per region for the 150-epoch U-Net and $\approx$20 for 80-epoch full-FT (Table 6; elapsed time was not retained in the result JSONs). The primary frozen-vs-fine-tuning comparison in Section 5 is separately budget-matched by construction (both 80 epochs, identical AdamW schedule and matched seeds; Table 10). Together with the 150-epoch convergence diagnostic (Appendix Figure 3), these controls do not indicate that the U-Net result is an artifact of under-training the GeoFMs.

External corroboration (PANGAEA). On AI4SmallFarms (Persello et al., 2023) (manually digitized field boundaries), PANGAEA (Marsocci et al., 2024) reports a from-scratch U-Net at 46.3 mIoU, beating every evaluated GeoFM (Prithvi 26.9, DOFA 27.1, Scale-MAE 21.5). Conversely, the Prithvi-EO-2.0 result of +8.1 mIoU over a from-scratch U-Net on multi-temporal crop segmentation (Szwarcman et al., 2024) comes from a crop-class segmentation task rather than a polygon-derived field-extent benchmark. Both evaluation choices appear in the published literature, independent of our evaluation setup.

5 Frozen Decoders Often Match Full Fine-tuning In-Region; Cross-Region Depends on Regime

Except for the frozen-probe boundary-zone control in Table 5, all GeoFM comparisons in this section use the frozen-decoder and full-FT configurations with matched training seeds; see Table 1. Holding the decoder, recipe, labels, and evaluation pixels fixed, Table 3 compares frozen-backbone GeoFM features against full end-to-end fine-tuning in a single-recipe ten-seed grid with ten matched seeds.

Table 3: **Frozen-decoder vs. full fine-tuning, in-region and cross-region (true-label AUROC).** *Left:* frozen backbone + trained decoder vs. full fine-tuning of the same architecture, with ten matched seeds per cell. 95% CIs on Δ = frozen − full-FT use a seed-paired t-interval (df = 9). A dagger marks TOST-equivalence at ε=0.02 AUROC by the conservative criterion that the 95% CI lies fully within $[-0.02, +0.02]$: 9 of 12 cells pass. Kenya rejects equivalence for both GeoFMs with full-FT ahead; TerraMind India is non-equivalent in the frozen-favored direction because the CI extends past +0.02. *Right:* mean AUROC over all 30 directed transfers and the exploratory non-Kenya 20 subset (see Section 5 for provenance). U-Net has the highest AUROC on the full matrix; on exploratory non-Kenya transfers, both frozen GeoFMs beat their full-FT counterparts, and the TerraMind-frozen-minus-U-Net AUROC CI includes zero.

In-region: frozen decoder vs. full-FT

Model	Region	frozen	full-FT	Δ	95% CI
Prithvi	India	0.982	0.985	−0.004	$[-0.006, -0.002]^\dagger$
Prithvi	Cambodia	0.947	0.949	−0.002	$[-0.003, -0.001]^\dagger$
Prithvi	Vietnam	0.955	0.958	−0.004	$[-0.004, -0.003]^\dagger$
Prithvi	Kenya	0.653	0.771	−0.117	$[-0.176, -0.059]$
Prithvi	France	0.982	0.979	+0.003	$[+0.002, +0.004]^\dagger$
Prithvi	Netherlands	0.971	0.958	+0.013	$[+0.008, +0.018]^\dagger$
TerraMind	India	0.968	0.948	+0.020	$[+0.005, +0.036]$
TerraMind	Cambodia	0.942	0.948	−0.006	$[-0.007, -0.005]^\dagger$
TerraMind	Vietnam	0.951	0.956	−0.004	$[-0.005, -0.004]^\dagger$
TerraMind	Kenya	0.613	0.710	−0.097	$[-0.146, -0.048]$
TerraMind	France	0.979	0.978	+0.001	$[+0.000, +0.002]^\dagger$
TerraMind	Netherlands	0.961	0.952	+0.009	$[+0.003, +0.015]^\dagger$

Cross-region mean AUROC

Config	all 30	expl. non-Kenya 20
U-Net (scratch)	**0.826**	**0.842**
TerraMind frozen-dec.	0.781	0.839
TerraMind full-FT	0.781	0.813
Prithvi full-FT	0.780	0.810
Prithvi frozen-dec.	0.767	0.816

† TOST-equivalent within ±0.02 AUROC (seed-paired t-interval, df = 9, n=10 seeds).

Table 4: **In-region IoU complements AUROC** (mean over 10 seeds, threshold p=0.5). The IoU pattern favors the U-Net more often than AUROC: U-Net wins outright in five regions (India, Vietnam, Kenya, France, Netherlands), while Full-FT leads in Cambodia. The Kenya entries for the frozen and full-FT configurations remain near all-negative at threshold 0.5. Full AUROC/AP/F1/IoU breakdown in Appendix Table 18.

Region (positive %)	U-Net IoU	Frozen-decoder IoU	Full-FT IoU
India (∼0.3%)	**0.455**	0.386	0.433
Cambodia (∼23%)	0.826	0.850	**0.856**
Vietnam (∼18%)	**0.854**	0.839	0.847
Kenya (∼0.1%)	**0.086**	0.000	0.005
France (∼11%)	**0.891**	0.850	0.856
Netherlands (∼3.8%)	**0.809**	0.699	0.708

In-region. With the backbone frozen and only a decoder trained, performance is statistically equivalent to full fine-tuning in 9 of 12 region×model cells across the single-recipe ten-seed grid (Table 3, TOST at ε=0.02 AUROC; count stable for $\varepsilon \in [0.02, 0.03]$; no FT-favored non-equivalence besides Kenya at any $\varepsilon \geq 0.010$; full per-seed breakdown in Appendix Table 8 and sensitivity in Appendix Table 9). Prithvi is equivalent outside Kenya: India, Cambodia, Vietnam, France, and the Netherlands. TerraMind is equivalent in four of six: Cambodia, Vietnam, France, and the Netherlands. The three non-equivalent cells have interpretable directions: Kenya favors full-FT for both GeoFMs (Prithvi Δ=−0.117, CI $[-0.176, -0.059]$; TerraMind Δ=−0.097, CI $[-0.146, -0.048]$), while TerraMind India favors the frozen decoder in direction but misses equivalence because the CI extends past the +0.02 margin (Δ=+0.020, CI $[+0.005, +0.036]$). Outside Kenya, the largest fine-tuning advantage is TerraMind Cambodia at 0.006 AUROC, well inside the declared equivalence band. An earlier apparent fine-tuning advantage came from an architecture confound: comparing a fine-tune-plus-decoder against a frozen linear probe. Giving the frozen backbone the same decoder removes the gap in most in-region cells.

IoU complements AUROC. For an imbalanced segmentation task (positive rates spanning 0.1%–23%), AUROC alone can obscure threshold behavior. Table 4 reports mean per-region IoU at threshold 0.5. The U-Net leads in five regions, while full-FT leads only in Cambodia. The Kenya entries for the frozen and full-FT configurations remain near all-negative at this threshold. Thus the segmentation metric favors the U-Net more often than AUROC does, but it does not show a systematic full-FT advantage over the frozen decoder. We do not make an IoU-equivalence claim.

Boundary-zone scope. Our task is field-extent segmentation, not boundary delineation or instance separation (Section 3). As a control on the frozen-probe comparisons in Table 2, we re-evaluated the frozen-probe feature classifiers using a boundary-zone pixel pool within ≤ 2 pixels of a field edge over the India chips (n=203,864), split 75/25 by chip. AUROC is computed only on the held-out split. The five classifiers evaluated here (four frozen probes + the per-pixel RF) all fall to near-chance on this slice (Table 5: AUROC 0.57–0.59 for Clay, Prithvi, TerraMind, AnySat, RF), against 0.94–0.98 on the full-region task for the three transformer-based FMs. We did not evaluate the U-Net, frozen-decoder, or full-FT configurations on this slice, so Table 5 supports *only* a frozen-probe scope claim: it does not rank the trained segmentation configurations on boundary pixels. Papers using FTW for true boundary delineation (e.g. object-F1 or Boundary-IoU style metrics) ask a strictly harder question and are outside our scope.

Table 5: **Frozen-probe boundary-zone evaluation (India).** A pool of n=203,864 pixels within ≤ 2 pixels of a field edge is split 75/25 by chip, and AUROC is computed only on the held-out split. The five classifiers evaluated here (four frozen probes + the per-pixel RF) all fall to near-chance AUROC, against 0.94–0.98 on the full-region task for the three transformer-based FMs. **Scope.** The U-Net, frozen-decoder, and full-FT configurations were *not* evaluated on this slice; this table supports only a frozen-probe scope claim and does not compare trained segmentation models on boundary pixels. None of the four frozen-probe configurations is a boundary-delineation specialist; the frozen-probe field-extent advantage rests on interior context.

Slice	RF	Clay	Prithvi	TerraMind	AnySat	full-region AUROC reference
boundary zone ($\leq$2 px)	0.586	0.571	0.576	0.584	0.582	see row below
full region (India)	0.574	0.577	0.983	0.967	0.939	cf. Table 2

Cross-region. For cross-country transfer without target-region labels, the ten-seed grid covers all 30 directed train-country→test-country transfers among the six regions (Appendix Table 11). The full matrix does *not* support a broad frozen-GeoFM transfer win: the from-scratch U-Net has the highest mean AUROC (0.826), followed by TerraMind full-FT and TerraMind frozen (both 0.781), Prithvi full-FT (0.780), and Prithvi frozen (0.767). Seed-paired deltas over the 30 transfers show U-Net ahead of both frozen GeoFMs (Prithvi frozen minus U-Net = -0.059, CI $[-0.067, -0.051]$; TerraMind frozen minus U-Net = -0.045, CI $[-0.053, -0.036]$). Prithvi frozen also trails Prithvi full-FT on the full matrix (-0.013, CI $[-0.021, -0.006]$), while the TerraMind comparison detects no difference (-0.000, CI $[-0.013, +0.013]$).

The non-Kenya subset was not pre-registered; we defined it after observing Kenya's in-region behavior and treat it as exploratory. The full 30-transfer matrix remains the confirmatory cross-region analysis. Kenya is independently characterized as a sparse-positive case on three grounds before its transfers are partitioned out: extreme label sparsity ($\sim$0.1% positive pixels, the lowest in the study), 155 training chips, and threshold-metric collapse on trained GeoFM configurations (Section 6, Appendix Table 18). On the exploratory non-Kenya 20 directed transfers, both frozen GeoFMs beat their full-FT counterparts: Prithvi frozen minus Prithvi full-FT is $+0.006$ AUROC (CI $[+0.001, +0.011]$), and TerraMind frozen minus TerraMind full-FT is $+0.025$ (CI $[+0.019, +0.032]$). Relative to the U-Net, the TerraMind frozen CI includes zero on exploratory non-Kenya transfers (-0.004, CI $[-0.013, +0.005]$), while Prithvi frozen is lower (-0.026, CI $[-0.033, -0.019]$). On the full matrix, a from-scratch U-Net has the highest AUROC and AP. The

IoU@0.5 ranking differs: TerraMind frozen exceeds the U-Net by 0.02 on the full matrix and by 0.05 on the exploratory subset (Appendix Table 12). The U-Net comparison is therefore specific to AUROC and AP. Under the thresholded segmentation metric, frozen TerraMind is competitive with or better than the U-Net. A task-trained supervised model also remains far stronger than a per-pixel spectral baseline (Table 2).

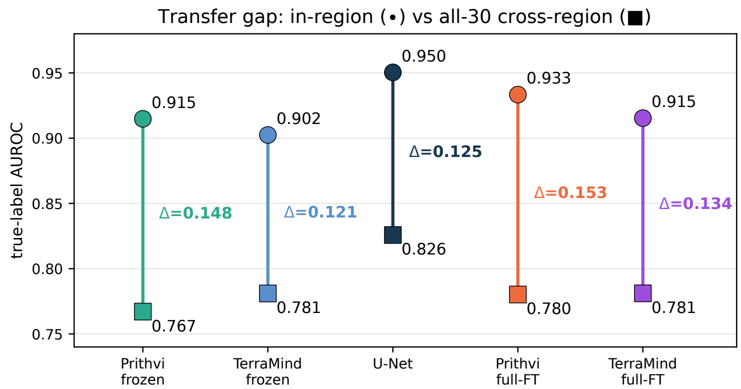

Figure 1: **Transfer gap.** Six-region in-region mean (●) vs. full 30-transfer cross-region mean (■). The U-Net has the highest AUROC on the full matrix; frozen decoders serve mainly as lower-cost alternatives to full fine-tuning, not as general cross-region winners.

6 Characterizing the Kenya Sparse-Positive Regime

The results in this section use the same frozen-decoder and full-FT configurations as Section 5; see Table 1. Kenya is the consistent sparse-positive case. In-region, full fine-tuning beats the frozen decoder for both GeoFMs in the ten-seed grid: Prithvi frozen 0.653 vs. full-FT 0.771 ($\Delta = -0.117$, CI $[-0.176, -0.059]$), and TerraMind frozen 0.613 vs. full-FT 0.710 ($\Delta = -0.097$, CI $[-0.146, -0.048]$). Cross-region, Kenya-involved transfers are also the hardest subset: on the 10 Kenya-involved directions, U-Net averages 0.793 AUROC, while frozen Prithvi and frozen TerraMind average 0.669 and 0.666. Two measured properties distinguish Kenya. Its true-field-pixel coverage in the training masks is about 0.1%, the lowest of any region; the next lowest is India at 0.3%. The released field-size summaries show similar shares of fields under 0.3 ha in India (65%) and Kenya (70%), yet only Kenya exhibits this collapse. Parcel size alone is therefore insufficient to explain the result. Cambodia has the highest positive-pixel coverage (23%) and remains stable. Kenya has 155 training chips, whereas Cambodia has 107 and remains stable, so training-set size alone is also insufficient. Spatial fragmentation may contribute, but we did not quantify it independently. Under this measured combination, frozen-feature decoders are unstable (seed standard deviation up to 0.066 AUROC), and the supervised U-Net is the most robust tested model on threshold metrics.

Reconciling Kenya numbers across tables. Kenya is the one region where the three configurations defined in Section 3 disagree numerically. **Best frozen-probe** (linear classifier on AnySat features, reported in Appendix Table 7 from a separate frozen-probe sensitivity, not the ten-seed decoder grid) scores 0.853 AUROC; **frozen-decoder** (Prithvi features, 2.75M-parameter decoder, Table 3) scores 0.653; the ten-seed threshold-metrics mean (Appendix Table 18) is 0.653 AUROC and 0.000 IoU. Tile-disjoint sensitivity (Appendix Table 17) further shows that AnySat's chip-grouped Kenya score does not survive a split-design change ($0.853 \rightarrow 0.580$), so this best-of-four Appendix cell is fragile in addition to being from a different configuration. One possible explanation is that the generic pretrained representation does not expose the features needed in this sparse regime. A frozen backbone cannot adapt those features, whereas full fine-tuning and a from-scratch U-Net can; they score 0.771 and 0.891 AUROC, respectively, compared with 0.653 for the frozen decoder. This is a hypothesis, not a causal result. It also argues against explaining the failure only by decoder size. In Section 7 and the cross-region narrative, we use the frozen-decoder Prithvi number (0.653) as the canonical Kenya frozen value, since it is the apples-to-apples comparator against fine-tuning. Threshold metrics sharpen the boundary: at probability threshold 0.5, the frozen decoder predicts all negatives and full fine-tuning is near all-negative on Kenya (mean IoU 0.000 and 0.005; Appendix Table 18); only the supervised U-Net recovers substantial positives (mean IoU 0.086). The Kenya failure appears in both AUROC and threshold metrics, so it is not an artifact of relying on AUROC alone.

We do not infer a hard sparsity threshold from one region. When annotated positives are very sparse and the training set is small, the results instead motivate a direct comparison between frozen decoders and a supervised CNN. Appendix Table 14 places Kenya in the context of the six regions.

A Cambodia subsampling control suggests that sample size alone does not explain the failure. Reducing its full-fine-tuning set to 11, 27, or 54 chips, all far below Kenya's 155, leaves AUROC at 0.940–0.949, compared with 0.949 on all 107 chips, and its thresholded F1 does not collapse (data in `ftw_finetune_fm_prithvi_camld_*.json`). Nor is the frozen-vs-fine-tuning difference monotone in positive-pixel prevalence across the other regions. The evidence therefore supports describing Kenya as a joint sparse-positive and small-training-set case, but it does not separate the two factors causally. A controlled study that varies positive prevalence while holding chip count fixed remains future work.

7 Cost-Efficiency of Frozen Features

Table 6 contrasts trainable parameters and per-region training cost. A Prithvi frozen backbone with a decoder trains 2.75M parameters versus 307M for Prithvi full fine-tuning, about 110× fewer. Across both GeoFMs, frozen decoders are TOST-equivalent to full fine-tuning in 9 of 12 region×model cells (Table 8). TerraMind India lies at the equivalence margin by point estimate (Δ=+0.020), but its CI extends beyond the margin. Kenya is the only region where full-FT decisively wins for both GeoFMs. A frozen linear probe trains about 10^3 parameters in seconds on CPU. AUROC parity with the 31M-parameter U-Net is region-dependent: in the ten-seed in-region decoder comparison, U-Net has higher point AUROC than frozen-Prithvi in four of six regions (Cambodia, Vietnam, Kenya, Netherlands), while frozen-Prithvi is higher in India and narrowly in France. The largest gap is Kenya, where U-Net leads frozen-Prithvi by 0.237 AUROC. On IoU, the U-Net wins outright outside Cambodia (Table 4). The frozen-probe cost advantage is robust; matched-accuracy equality vs. the supervised baseline is region-specific. For cross-region deployment, the frozen decoder also needs no target-region labels, but the full transfer matrix shows that it should be compared directly with a supervised U-Net; its transfer strength cannot be assumed.

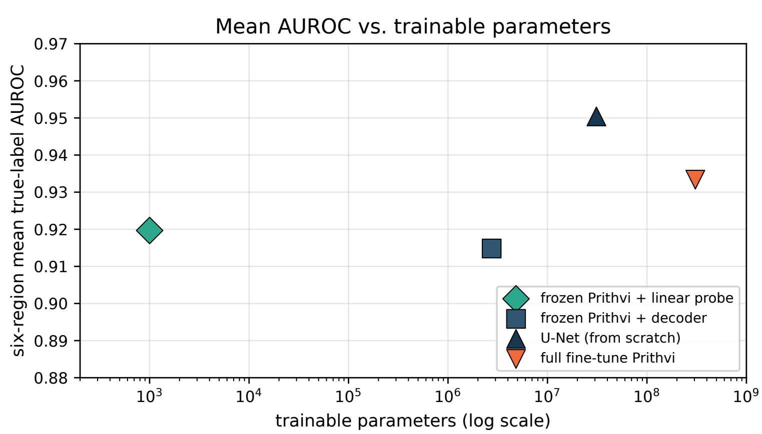

Figure 2: **Cost vs. performance.** Six-region mean in-region AUROC against trainable parameters (log scale). All GeoFM points use Prithvi consistently. Frozen options (a ∼1K-parameter linear probe or a 2.75M-parameter decoder) sit far left of the fully trained models. The frozen *decoder* has 2.75M trainable parameters for Prithvi, compared with 307M for Prithvi full-FT. Across both GeoFMs, frozen decoders are TOST-equivalent to full-FT in 9 of 12 region×model cells (Table 8). The frozen *probe* has lower point AUROC than the 31M from-scratch U-Net in most regions; India is the main GeoFM exception.

Table 6: **Cost asymmetry.** Trainable parameters and per-region training cost. The frozen-decoder pipeline is TOST-equivalent to full fine-tuning in 9 of 12 region×model cells (Table 8). For Prithvi, the frozen decoder trains a small fraction of the parameters used by full fine-tuning. GPU-minute entries are approximate development-run observations; elapsed time was not retained in the result JSONs. AUROC parity with the from-scratch U-Net is region-dependent; on IoU, the U-Net wins outright outside Cambodia (Table 4).

Method	trainable params	per-region train	target labels
per-pixel RF	N/A (fitted trees)	seconds, CPU	yes
frozen FM + linear probe	∼1,000	seconds, CPU	yes (source only, transfer)
frozen Prithvi + decoder	2.75M	∼12 GPU-min	yes
supervised U-Net	31M	∼15 GPU-min	yes
full fine-tune (Prithvi)	307M	∼20 GPU-min	yes

Cost accounting: what is and is not counted. Table 6 reports trainable parameters and approximate per-region wall-clock observations from a single A6000 GPU. Because elapsed time was not stored in the result JSONs, these timings indicate scale but are not part of the reproducible quantitative claims. The auditable cost statements are precisely scoped to two facts. First, the Prithvi frozen decoder trains ∼110× fewer parameters than Prithvi full-FT (2.75M vs. 307M). Second, for both GeoFMs, the frozen-decoder and full-FT configurations run the identical backbone+decoder graph at inference, so per-scene inference cost is equal; frozen does not reduce deployment-time backbone compute. The released frozen-decoder pipeline does not cache features offline: the frozen backbone still runs a forward pass every training epoch, so there is no per-epoch compute saving from cached features and no cached-feature storage term to report. A user who chooses to cache frozen features offline would pay a one-time extraction and storage cost and could then reuse those features across multiple task heads, but our released pipeline does not do this and therefore does not realize that multi-head-reuse saving.

8 A Corrected-Evaluation Protocol

We distill the study into a ten-point checklist for field-level GeoFM benchmarking.

1. **Labels.** Evaluate against field-extent labels derived from field polygons, not a land-cover proxy; treat proxy-only rankings as tentative.
2. Compare label sources on the same pixels.
3. **Baselines.** Include a strong task-specific supervised baseline such as a U-Net alongside any per-pixel spectral model.
4. Report both a frozen-feature configuration and a fine-tuned variant; test whether fine-tuning helps.
5. **Splits.** Group examples by chip.
6. Add a scene- or tile-disjoint split and a cross-region split.
7. **Preprocessing.** Use each model's own normalization and exclude no-data pixels.
8. **Statistics.** Cluster uncertainty at the chip level, use multiple seeds, and correct for multiple comparisons.
9. **Interpretation.** Run an edge-vs-interior control before claiming boundary understanding.
10. Report field-size and label-sparsity regimes, trainable parameters, and compute cost alongside accuracy. In this study, Kenya is the only region below approximately 0.2% positive pixels; this is a descriptive warning regime rather than a general threshold.

Worked example: re-reading a published ranking. The Prithvi-EO-2.0 report (Szwarcman et al., 2024) reports a +8.1 mIoU gap for Prithvi-EO-2.0-600M over a from-scratch U-Net on multi-temporal crop segmentation (50.7 vs. 42.6 mIoU in its Table 9). Protocol step (1) flags direct transfer of that ranking to our setting as task-dependent: crop-class segmentation is not polygon-derived field-extent segmentation, so the result should be re-tested on field-polygon labels before being treated as evidence about field membership. With polygon-derived labels and a strong baseline (steps 1 to 3), PANGAEA (Marsocci et al., 2024) reports on AI4SmallFarms (Persello et al., 2023) (manually digitized parcels) a from-scratch U-Net at 46.3 mIoU, beating every evaluated GeoFM (Prithvi 26.9, DOFA 27.1, Scale-MAE 21.5). Multi-temporal crop segmentation and single-date AI4SmallFarms parcel delineation are different tasks/datasets, so this is not a direct apples-to-apples inversion of the Prithvi result; rather, it illustrates why proxy-label wins should be re-tested under steps (1)–(3).

Practitioner implications. For Prithvi, a frozen backbone with a decoder trains about 110× fewer parameters than full fine-tuning; a linear probe uses roughly 10^3 trainable parameters (Table 6). The released frozen-decoder pipeline does not cache features: it still runs the backbone forward pass during training, and it uses the same backbone+decoder graph as full-FT at inference. Users who explicitly cache features offline can trade one-time extraction and storage for multi-head reuse, but that saving is not realized in the reported runs.

For a new region with no local labels, the exploratory non-Kenya transfer subset supports frozen decoders over full fine-tuning for both GeoFMs, but the full cross-region matrix does not support a broad claim that frozen GeoFMs beat a source-trained U-Net. On the full 30-transfer matrix the U-Net has the highest AUROC and AP; on exploratory non-Kenya transfers no AUROC difference is detected between TerraMind frozen and the U-Net, while Prithvi frozen is lower.

For this single-date task, a frozen decoder is a low-cost configuration to test first, not a general replacement for full fine-tuning or a substitute for a strong supervised baseline; sparse-positive regimes require a direct comparison with a supervised CNN.

9 Discussion, Limitations, Conclusion

Scope. Our claims are scoped to agricultural field-extent segmentation from single-date Sentinel-2 with true field labels (per-pixel inside-vs-outside field polygon; we do not evaluate boundary delineation or instance separation). They do not necessarily extend to other tasks (e.g. crop-type, change detection), other sensors, or multi-temporal regimes. GeoFM pretraining may help there.

Broader impact. Automatic field-extent maps are used downstream in agricultural planning, subsidy allocation, crop monitoring, and land-tenure assessment. The same maps can also enable land-use surveillance, regulatory enforcement, or asset valuation that affects smallholder farmers without their consent. Our methodology choices (polygon-derived labels instead of land-cover proxies, a strong supervised baseline, explicit reporting of boundary-zone failure) aim to keep claims about "what works" aligned with the actual task and to make weakness in sparse-label regimes (e.g. Kenya) explicit. Deployers should not generalize results across agro-climatic regimes without re-evaluating on local labels, and decisions with consequences for individuals should not be automated from these maps alone.

Scope of the claim. The claim is not that GeoFMs are useless. Current evaluations overstate their advantage for in-distribution field mapping with reasonable supervision; the practical value here is as frozen feature extractors, which are low-cost and often match full fine-tuning in-region. Cross-region, the value is narrower: frozen decoders beat full fine-tuning on the exploratory non-Kenya transfer subset, but the U-Net has the highest AUROC and AP on the full directed matrix.

Limitations. Our model coverage includes four GeoFMs and one supervised architecture. PANGAEA evaluates more GeoFMs and reports a similar U-Net pattern, but it does not replace broader testing within our setup. FTW annotations can also be partial in sparse regions. A three-region sensitivity check does not identify that partial annotation as the source of the frozen-FM result (Appendix Table 16), but it cannot rule out every labeling effect. Finally, the cross-region AUROC evidence is deliberately narrow: the U-Net leads the full 30-transfer matrix, and the statistically supported frozen-over-FT result is limited to the exploratory non-Kenya subset.

Training budgets. The U-Net-vs-GeoFM comparisons are architecture-appropriate but not compute-matched: the U-Net uses Adam for 150 epochs from random initialization while the GeoFM frozen-decoder and full-FT runs use AdamW for 80 epochs from pretrained weights. An 80-epoch U-Net control (Appendix Table 13, 3 seeds per region) preserves the qualitative pattern of Table 2 within 0.01 AUROC in five of six regions; India shows a 0.03 drop that widens the existing India frozen-probe lead but does not create a new region-level exception. The frozen-decoder-vs-full-FT primary comparison is separately budget-matched by construction (Table 10). Under our approximate development-run timings, the U-Net is cheaper than full-FT per region despite its longer schedule, although elapsed time was not retained in the result JSONs. A per-epoch convergence diagnostic (Appendix Figure 3) shows that the Vietnam frozen-decoder curve changes by about 0.002 AUROC from epoch 15 to 150, while the Vietnam full-FT curve peaks near epoch 90 and then declines slightly. The Kenya full-FT curve peaks at epoch 15 and ends lower at epoch 150, despite transient rebounds. No configuration shows a sustained gain beyond the 80-epoch budget.

Boundary and cost controls. The boundary-zone control (Table 5) is limited to the frozen-probe configurations; extending it to the U-Net, frozen-decoder, and full-FT configurations would require regenerating per-pixel predictions for those trained models. We therefore report Table 5 strictly as a frozen-probe scope check. Feature-extraction wall-time is not separately tabulated because the frozen-decoder configuration used in the frozen-vs-fine-tuning spine does not cache features to disk; extraction is folded into the training wall-time (Table 6). The table's GPU-minute values are approximate, unlogged observations; the auditable cost comparison is the trainable-parameter count.

Connection to the broader foundation-model literature. For geospatial models, the cross-region result is partly consistent with what Kumar et al. (2022) showed for vision: fine-tuning can distort pretrained features and underperform frozen probing under distribution shift. We see that pattern on the exploratory non-Kenya transfer subset, where frozen decoders beat full fine-tuning for both GeoFMs. The full 30-transfer matrix is more cautious because Kenya-involved transfers favor task-trained supervised models and the U-Net has the highest AUROC and AP overall. This supports a narrower recommendation: test a frozen decoder before full fine-tuning, but do not treat it as a transfer rule. The released evaluation setup can test where this pattern persists.

Conclusion. Two evaluation choices materially change this field-extent benchmark. WorldCover cropland makes the target more spectrally separable than polygon-derived field extent, and a task-trained U-Net is much stronger than a per-pixel random forest. GeoFM rankings should therefore be interpreted only after the label and baseline choices are made explicit.

In the controlled ten-seed comparison, frozen decoders are TOST-equivalent to full fine-tuning in 9 of 12 region×model cells. For Prithvi, the frozen-decoder configuration trains about 110× fewer parameters. Kenya is the shared failure case for both GeoFMs. Cross-region, the U-Net leads the full 30-transfer matrix on AUROC and AP. On AUROC, frozen decoders outperform full fine-tuning only in the exploratory subset that excludes Kenya, and no difference is detected between frozen TerraMind and the U-Net in AUROC there. The practical recommendation is to test a frozen decoder before paying for full fine-tuning, while retaining a strong supervised baseline and checking sparse-positive regions directly. The released code, metrics, and protocol support the same audit in other GeoFM evaluations. These conclusions apply to single-date, per-pixel field-extent segmentation, not parcel-boundary or instance delineation; multi-temporal regimes, where Prithvi-EO-2.0 was designed to operate, are out of scope.

Acknowledgments

The author thanks Knox College for providing the computational resources used in this work. This research received no external funding.

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A Reproducibility and Controls

This appendix provides the detailed results and controls referenced in the main text. Tables 7–10 cover the frozen-probe breakdown, paired TOST analysis, margin sensitivity, and training recipes. Tables 11–12 report the full transfer matrix and its AP/IoU companion. The remaining tables and Figure 3 report budget, regional, spatial, partial-label, split, and threshold-metric controls. Unless a caption states otherwise, values are generated by the released scripts from the released result files; rounded mask-prevalence descriptors are recorded in `scripts/integrate_revision_controls.py` because source masks are not redistributed.

Table 7: **Per-region breakdown of frozen-probe AUROC across four FMs** (true-label, linear probe, deterministic given the split). The Table 2 main column uses Prithvi-EO-2.0 consistently; this table reports each FM separately for transparency. **Bold** marks the per-region winner. AnySat wins Kenya (0.853) and Netherlands (0.961), but tile-disjoint sensitivity (Table 17) shows the AnySat scores do not survive a split-design change (0.853 → 0.580 on Kenya; 0.943 → 0.605 on France); we therefore use the Prithvi-only column as the main comparison. Bootstrap 95% CIs from B=1000 chip-level resamples.

Region	Prithvi	TerraMind	Clay	AnySat
India	**0.983** [.979,.985]	0.967 [.963,.971]	0.577 [.554,.601]	0.939 [.915,.959]
Vietnam	**0.929** [.925,.933]	0.916 [.911,.921]	0.503 [.495,.511]	0.897 [.880,.915]
Kenya	0.766 [.718,.813]	0.745 [.699,.789]	0.556 [.505,.599]	**0.853** [.825,.881]
France	**0.961** [.958,.964]	0.951 [.947,.954]	0.501 [.494,.509]	0.943 [.927,.957]
Netherlands	0.955 [.951,.958]	0.943 [.938,.948]	0.498 [.488,.507]	**0.961** [.956,.967]

Cambodia per-chip bootstrap CIs were not recomputed; available per-model point estimates are Prithvi 0.924, TerraMind 0.918, and Clay 0.491. No Cambodia AnySat result is present in the artifact.

Table 8: **Seed-paired TOST equivalence for frozen-decoder vs. full fine-tune (12 cells: two GeoFMs × six regions).** Per-seed AUROC values (seeds 0–9), paired delta Δ_s = frozen$_s$ − FT$_s$ averaged over seeds, 95% paired t-interval with df = 9, and TOST verdict at declared equivalence margin ε=0.02. The paired analysis uses matched random seeds for each frozen/FT pair in the single-recipe 10-seed grid. Source files and aggregate JSON are generated by `scripts/integrate_headline_results.py`. Total TOST-equivalent: 9 of 12.

Model	Region	frozen seeds (s0–s9)	FT seeds (s0–s9)	paired $\Delta \pm \sigma$	95% CI
Prithvi	India	.984, .981, .983, .985, .983, .979, .980, .979, .985, .978	.982, .988, .985, .986, .986, .986, .983, .982, .988, .986	−0.004 ± 0.003	[−0.006, −0.002] ✓
Prithvi	Cambodia	.948, .948, .946, .947, .946, .946, .948, .948, .948, .947	.949, .949, .948, .950, .949, .949, .949, .949, .949, .948	−0.002 ± 0.001	[−0.003, −0.001] ✓
Prithvi	Vietnam	.954, .955, .954, .955, .955, .955, .955, .954, .955, .955	.958, .957, .958, .959, .959, .959, .958, .958, .958, .960	−0.004 ± 0.001	[−0.004, −0.003] ✓
Prithvi	Kenya	.667, .658, .615, .599, .607, .676, .531, .723, .745, .713	.781, .798, .699, .811, .814, .728, .761, .701, .791, .824	−0.117 ± 0.082	[−0.176, −0.059] ×
Prithvi	France	.981, .982, .981, .982, .982, .982, .982, .981, .981, .981	.980, .978, .981, .979, .979, .977, .977, .979, .979, .979	+0.003 ± 0.002	[+0.002, +0.004] ✓
Prithvi	Netherlands	.971, .973, .971, .971, .970, .974, .971, .968, .968, .971	.964, .957, .953, .954, .963, .961, .946, .964, .951, .965	+0.013 ± 0.007	[+0.008, +0.018] ✓
TerraMind	India	.972, .934, .976, .977, .973, .975, .978, .964, .970, .965	.970, .955, .950, .946, .961, .956, .946, .930, .912, .955	+0.020 ± 0.021	[+0.005, +0.036] ×
TerraMind	Cambodia	.943, .942, .943, .942, .941, .943, .943, .942, .942, .942	.949, .948, .947, .949, .949, .948, .949, .949, .949, .947	−0.006 ± 0.001	[−0.007, −0.005] ✓
TerraMind	Vietnam	.951, .951, .952, .951, .951, .951, .952, .951, .951, .951	.955, .956, .956, .956, .956, .956, .956, .956, .955, .956	−0.004 ± 0.000	[−0.005, −0.004] ✓
TerraMind	Kenya	.610, .632, .638, .596, .537, .576, .707, .643, .569, .625	.820, .724, .788, .611, .573, .696, .696, .773, .721, .703	−0.097 ± 0.069	[−0.146, −0.048] ×
TerraMind	France	.978, .979, .979, .979, .979, .979, .979, .979, .979, .978	.979, .978, .978, .978, .977, .979, .975, .978, .978, .978	+0.001 ± 0.001	[+0.000, +0.002] ✓
TerraMind	Netherlands	.960, .958, .958, .956, .964, .965, .964, .960, .964, .959	.942, .951, .961, .953, .943, .964, .962, .940, .954, .952	+0.009 ± 0.009	[+0.003, +0.015] ✓

✓ = TOST equivalent at ε=0.02 AUROC (95% paired CI fully within ±0.02; equivalent to standard TOST at α=0.025 per side, conservative form). × = not formally equivalent. Kenya is a full-FT win for both GeoFMs; TerraMind India is frozen-favored but its CI extends past +0.02.

Table 9: **Equivalence sensitivity to the TOST margin.** Table counts use the same 95% seed-paired CI-containment criterion as the primary verdicts (df = 9). At five of the six margins, Holm correction across the 12 TOST p-values retains the same number of cells. At ε=0.015, it retains 7 rather than 8 because TerraMind–Netherlands has p_{Holm}=0.112. The failing set is invariant across $\varepsilon \in [0.020, 0.030]$. Cells added at stricter margins in $[0.010, 0.020)$ are frozen-favored (F), so tightening the margin in this range only reclassifies cells where the frozen decoder already beats fine-tuning. At ε=0.005 two additional cells become non-equivalent in the FT-favored direction (T), but their Δ magnitudes (0.004 and 0.006) are below the declared practical tolerance.

ε	Equivalent	Newly failing (direction, Δ)	Failing cells at this ε
0.030	9/12	none	{Pr–Kenya (T), TM–India (F), TM–Kenya (T)}
0.025	9/12	none	{same}
0.020	9/12	none	{same}
0.015	8/12	Pr–Netherlands (F, +0.013)	{above} $\cup$ Pr–Netherlands
0.010	7/12	TM–Netherlands (F, +0.009)	{above} $\cup$ TM–Netherlands
0.005	5/12	Pr–India (T, −0.004); TM–Cambodia (T, −0.006)	{above} $\cup$ Pr–India, TM–Cambodia

All experiments use the same chip-grouped split (seed 20260514; no chip overlap). Primary results recompute from the released run files, and direct raster checks verify label-to-image alignment in all six regions. The released controls cover spatial pooling, frozen-probe edge behavior, partial labels, and tile-disjoint splits. The single-recipe ten-seed grid includes all 30 directed transfers for U-Net, Prithvi, and TerraMind. Historical pilot files are retained for traceability but are not used for the paper's claims. Code, checksums, reproduction commands, and Croissant metadata are included in the supplement and at `https://github.com/thesantoshpant/auditing-geofm-evaluation` and archived at `https://doi.org/10.5281/zenodo.21993508` (v1.0.0).

Table 10: **Training recipes and controlled factors.** Frozen-decoder and full-FT GeoFM runs share the same decoder, loss, augmentation, schedule, and matched seeds; they differ only in whether the backbone is updated. The U-Net uses its architecture-appropriate optimizer and longer from-scratch schedule. Evaluation pixels, labels, and the chip-grouped split are identical for every model and seed. The U-Net and GeoFM training budgets differ by design (150 ep from-scratch vs. 80 ep from pretrained), matching common practice for each architecture. An 80-epoch U-Net sensitivity (Appendix Table 13) and a per-epoch convergence diagnostic (Appendix Figure 3) are provided for audit purposes.

Component	Setting
Loss	BCE (capped positive weight, ≤ 50) + Dice
Optimizer	U-Net: Adam, lr 10^{-3}; GeoFM: AdamW (backbone 10^{-4}, decoder 10^{-3})
LR schedule	linear warmup (5 epochs, start factor 0.1) + cosine decay
Augmentation	random horizontal/vertical flips
Input	U-Net: native chip, 12 S2 bands (reflect-padded); GeoFM: 224×224
Decoder	GeoFM: shared 4-stage bilinear-upsample conv decoder (identical for frozen & fine-tuned); U-Net: 4-level (31M)
Epochs	U-Net 150 (from scratch); GeoFM full-FT / frozen-decoder 80
Selection	last epoch (no early stopping); mean±std over 10 matched training seeds in the ten-seed grid

Table 11: **Full cross-region transfer matrix** (true-label AUROC, mean over 10 seeds; train country A, test country B's canonical test pixels). The table covers all 30 directed transfers among the six regions. U-Net has the highest mean AUROC on the full matrix (0.826). On the exploratory non-Kenya 20 subset, TerraMind frozen-decoder (0.839) is close to U-Net (0.842) and beats TerraMind full-FT (0.813), while Prithvi frozen-decoder (0.816) beats Prithvi full-FT (0.810) but remains below U-Net. Seed-paired CIs for subset-level deltas are generated in `data/results/ftw_cross_region_transfer.json`. In the compact headers, "frz" denotes frozen-decoder.

A→B	U-Net	Prithvi-FT	Prithvi-frz	TerraMind-FT	TerraMind-frz
India→Cambodia	0.719	0.646	0.629	0.627	0.697
India→Vietnam	0.698	0.664	0.653	0.643	0.699
India→Kenya	0.696	0.629	0.579	0.546	0.616
India→France	0.824	0.750	0.644	0.611	0.660
India→Netherlands	0.830	0.746	0.675	0.614	0.646
Cambodia→India	0.832	0.861	0.890	0.870	0.885
Cambodia→Vietnam	0.874	0.926	0.922	0.940	0.917
Cambodia→Kenya	0.881	0.836	0.865	0.862	0.853
Cambodia→France	0.865	0.887	0.910	0.920	0.884
Cambodia→Netherlands	0.895	0.883	0.884	0.901	0.879
Vietnam→India	0.873	0.868	0.870	0.875	0.886
Vietnam→Cambodia	0.939	0.938	0.938	0.945	0.930
Vietnam→Kenya	0.897	0.862	0.884	0.890	0.885
Vietnam→France	0.925	0.907	0.919	0.930	0.931
Vietnam→Netherlands	0.884	0.843	0.898	0.898	0.898
Kenya→India	0.790	0.678	0.517	0.642	0.596
Kenya→Cambodia	0.884	0.723	0.622	0.734	0.592
Kenya→Vietnam	0.746	0.642	0.588	0.690	0.543
Kenya→France	0.800	0.767	0.629	0.692	0.550
Kenya→Netherlands	0.851	0.770	0.640	0.738	0.596
France→India	0.887	0.877	0.903	0.891	0.904
France→Cambodia	0.840	0.740	0.748	0.781	0.750
France→Vietnam	0.835	0.688	0.742	0.794	0.828
France→Kenya	0.761	0.672	0.722	0.714	0.735
France→Netherlands	0.937	0.905	0.931	0.883	0.926
Netherlands→India	0.864	0.856	0.898	0.871	0.881
Netherlands→Cambodia	0.691	0.700	0.691	0.716	0.777
Netherlands→Vietnam	0.692	0.619	0.627	0.648	0.834
Netherlands→Kenya	0.625	0.618	0.642	0.663	0.692
Netherlands→France	0.940	0.901	0.954	0.904	0.960
mean all 30	0.826	0.780	0.767	0.781	0.781
mean expl. non-Kenya 20	0.842	0.810	0.816	0.813	0.839

Table 12: **Cross-region AP and IoU alongside AUROC (mean over 10 seeds per pair, then mean over pairs).** Threshold metrics (IoU at 0.5) and AP are reported in addition to AUROC because low positive rates (Kenya $\sim 0.1\%$, India $\sim 0.3\%$) can make AUROC alone optimistic. The principal AUROC conclusion is preserved under AP: U-Net leads the full matrix, and U-Net and TerraMind frozen-decoder differ by only 0.001 AP on the exploratory non-Kenya subset. The IoU@0.5 column is a thresholded segmentation-quality metric; under this metric the ordering is metric-dependent (see Section 5). Aggregated from existing per-run JSONs; no new experiments. See `data/results/ftw_xfer_metrics.json`.

	all 30 directed transfers			exploratory non-Kenya 20		
Config	AUROC	AP	IoU@0.5	AUROC	AP	IoU@0.5
U-Net (scratch)	**0.826**	**0.733**	0.335	**0.842**	**0.805**	0.423
TerraMind frozen-dec.	0.781	0.696	**0.353**	0.839	0.804	**0.469**
TerraMind full-FT	0.781	0.693	0.318	0.813	0.778	0.416
Prithvi full-FT	0.780	0.689	0.308	0.810	0.771	0.402
Prithvi frozen-dec.	0.767	0.682	0.308	0.816	0.782	0.404

Table 13: **U-Net epoch sensitivity.** The 80-epoch control reports mean±standard deviation over seeds 0–2; the 150-epoch column reports the primary ten-seed mean. The comparison is a budget sensitivity check, not a seed-paired test; architecture, optimizer, data, and schedule family are unchanged. The displayed Δ is computed from unrounded means. At 80 epochs the U-Net remains within 0.01 AUROC of the 150-epoch mean in five of six regions; India shows a 0.03 drop but remains the same region-level exception by point estimate. Training-time cost per region is roughly proportional to epochs (Table 6).

Region	U-Net@150ep	U-Net@80ep	Δ AUROC	Prithvi frozen-probe
India	0.941	0.911 ± 0.023	-0.030	0.983
Cambodia	0.948	0.947 ± 0.001	-0.001	0.924
Vietnam	0.964	0.955 ± 0.001	-0.009	0.929
Kenya	0.891	0.889 ± 0.004	-0.002	0.766
France	0.977	0.976 ± 0.001	-0.001	0.961
Netherlands	0.981	0.982 ± 0.003	$+0.000$	0.955

Table 14: **Per-region operational regime (descriptive, existing artifacts only).** Positive-pixel prevalence and training-set size vary by orders of magnitude across the six regions. Combined with the frozen-decoder-vs-full-fine-tuning ΔAUROC (Prithvi and TerraMind) and threshold-metric behavior (U-Net vs. frozen-decoder IoU@0.5), this shows where frozen features remain competitive and where they fail in these six regions. Kenya is the only region below $\sim 0.2\%$ positive pixels, so we do not report a formal sparsity threshold; the pattern is a description, not an estimate. Sample-size alone is not the mechanism: sub-sampling Cambodia (dense-positive) full-fine-tuning to 11–54 training chips leaves AUROC essentially flat (0.940–0.949; `ftw_finetune_fm_prithvi_camld_*`). A controlled frozen-decoder low-data arm at Kenya-scale chip count is future work.

Region	pos. %	train chips	ΔAUROC Pr	ΔAUROC TM	U-Net IoU	Frozen IoU	Regime
Kenya	$\sim$0.1	155	-0.117	-0.097	0.086	0.000	sparse-positive collapse
India	$\sim$0.3	525	-0.004	$+0.020$	0.455	0.386	smallholder; model-dependent
Netherlands	$\sim$3.8	522	$+0.013$	$+0.009$	0.809	0.699	frozen-favored AUROC
France	$\sim$11	532	$+0.003$	$+0.001$	0.891	0.850	industrial; near-parity
Vietnam	$\sim$18	213	-0.004	-0.004	0.854	0.839	tie
Cambodia	$\sim$23	107	-0.002	-0.006	0.826	0.850	tie

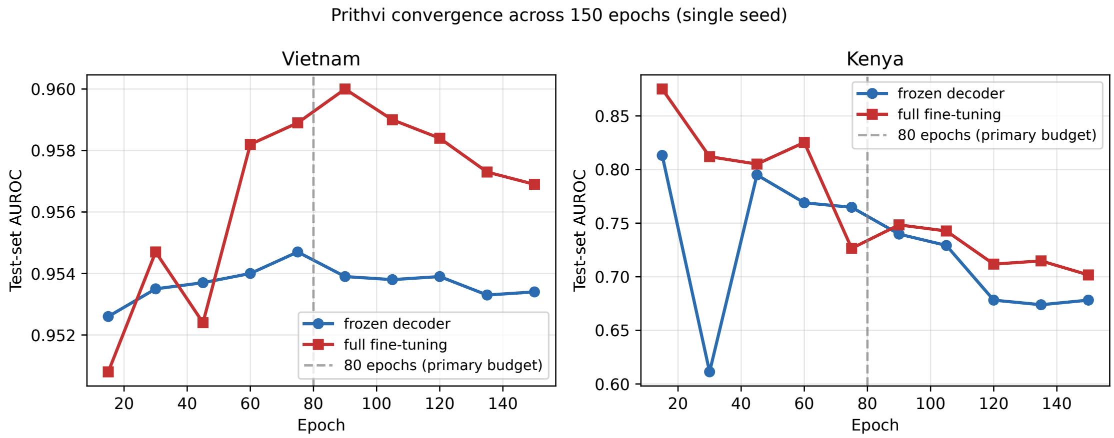

Figure 3: **Prithvi test-set AUROC across 150 epochs (single-seed diagnostic).** Post-hoc test-set AUROC for Prithvi frozen-decoder and full fine-tuning on Vietnam (dense-positive, 18% positives) and Kenya (sparse-positive case, 0.1% positives). Vertical dashed line at epoch 80 marks the primary-comparison budget. *Vietnam:* the frozen-decoder curve changes by about 0.002 AUROC from epoch 15 through 150; the full-FT curve peaks near epoch 90 (≈0.960) and slightly declines to 0.957 at epoch 150. *Kenya:* the full-FT curve peaks at epoch 15 (0.875) and declines to 0.702 by epoch 150, with small transient rebounds around epochs 60 and 90; the frozen-decoder is unstable and never recovers a stable value in this regime, ending at 0.678. No configuration shows a sustained gain beyond the 80-epoch budget, and both Kenya curves end lower at epoch 150 despite transient rebounds. These post-hoc trajectories were not used to choose checkpoints or hyperparameters; model selection remains last-epoch with no early stopping.

Table 15: **Spatial-pooling control (true-label AUROC).** A 5×5 windowed random forest (300-d) gains only about 0.01 over the per-pixel RF on average (Kenya drops by the same amount) and stays far below the best foundation model. This supports the interpretation that the GeoFM and U-Net advantage uses learned spatial context, which naive local pooling lacks. A per-pixel U-Net ablation (1×1 convolutions, no spatial context) likewise falls to ≈RF (India 0.623, Vietnam 0.678).

Region	per-pixel RF	windowed RF (5×5)	best FM
India	0.574	0.582	0.983
Vietnam	0.643	0.657	0.929
Kenya	0.651	0.641	0.853
France	0.695	0.703	0.961
Netherlands	0.815	0.827	0.961

Table 16: **FTW partial-labeling sensitivity (true-label AUROC).** Restricting negatives to high-confidence non-cropland (dropping ambiguous, possibly-unlabeled field pixels) leaves the frozen-GeoFM nearly unchanged in the three tested regions, while the spectral RF rises. This control does not indicate that FTW partial annotation drives the frozen-GeoFM result. France's RF gain ($0.695 \rightarrow 0.947$) reflects spectrally crop-like negatives (pasture, vineyards) that confuse a per-pixel RF; removing them leaves an easy task. The GeoFM was near-ceiling, so it barely moves.

Region	RF (all neg)	RF (high-conf)	Prithvi (all neg)	Prithvi (high-conf)
India	0.574	0.665	0.983	0.987
France	0.695	0.947	0.961	0.968
Kenya	0.651	0.698	0.766	0.774

Table 17: **Tile-disjoint sensitivity for the Table 2 frozen-probe column (true-label AUROC).** Each pair: chip-grouped (used in the main text) vs. MGRS-tile-disjoint AUROC for the same frozen probe. Prithvi-EO-2.0 is stable across both splits (drops ≤ 0.007). TerraMind is also stable except in Kenya, where it drops from 0.745 to 0.698. AnySat drops on Kenya (0.853 $\rightarrow$ 0.580), Vietnam (0.897 $\rightarrow$ 0.645), and France (0.943 $\rightarrow$ 0.605): its chip-grouped advantage may not reflect transferable feature quality. In Kenya, Prithvi (0.761) is more robust under tile-disjoint than AnySat (0.580). Cambodia is omitted (variant not run). Chip-grouped numbers preserve the per-chip statistical assumptions of the controlled comparison; this is the sensitivity check.

	Prithvi-EO-2.0		TerraMind		AnySat	
Region	chip	tile	chip	tile	chip	tile
India	0.983	0.984	0.967	0.969	0.939	0.945
Vietnam	0.929	0.922	0.916	0.915	0.897	0.645
Kenya	0.766	0.761	0.745	0.698	0.853	0.580
France	0.961	0.959	0.951	0.947	0.943	0.605
Netherlands	0.955	0.955	0.943	0.944	0.961	0.893

Table 18: **Threshold-based metrics complement AUROC** (mean over 10 seeds; F1/IoU at p=0.5; AP threshold-free). Dense-positive Cambodia and Vietnam show the three configurations close on IoU. France and Netherlands show the U-Net ahead on IoU, so the threshold metrics favor the supervised baseline more than AUROC alone would suggest. Kenya ($\sim$0.1% positives): frozen decoder predicts all negatives (F1=IoU=0), full fine-tune is near all-negative, and U-Net retains AP=0.593, F1=0.159. "Frozen" = Prithvi backbone frozen, decoder trained.

Region	Model	AUROC	AP	F1	IoU
India	U-Net	0.941	0.856	0.619	0.455
India	Frozen	0.982	0.970	0.557	0.386
India	Full-FT	0.985	0.974	0.604	0.433
Cambodia	U-Net	0.948	0.923	0.904	0.826
Cambodia	Frozen	0.947	0.912	0.919	0.850
Cambodia	Full-FT	0.949	0.914	0.923	0.856
Vietnam	U-Net	0.964	0.948	0.921	0.854
Vietnam	Frozen	0.955	0.927	0.912	0.839
Vietnam	Full-FT	0.958	0.933	0.917	0.847
Kenya	U-Net	0.891	0.593	0.159	0.086
Kenya	Frozen	0.653	0.304	0.000	0.000
Kenya	Full-FT	0.771	0.403	0.010	0.005
France	U-Net	0.977	0.963	0.942	0.891
France	Frozen	0.982	0.972	0.919	0.850
France	Full-FT	0.979	0.972	0.923	0.856
Netherlands	U-Net	0.981	0.977	0.895	0.809
Netherlands	Frozen	0.971	0.967	0.823	0.699
Netherlands	Full-FT	0.958	0.962	0.829	0.708